1/1/26 Recitation 1

) A smart farming system predicts whether a plant will grow well (1) or poorly (0) based on daily sunlight hours & water amount in litres.

Sunlight Hours (x_1)	Water in litres (x_2)	Bias (x_3)
3	1	1
8	4	1
5	2	1
10	5	1
7	3	1

Target (Growth = 1 / Poor = 0) (Y)
0
1
0
1
1

Mean of x_1 => $\mu x_1 = \frac{3+8+5+10+7}{5} = \frac{33}{5}$

$= 6.6$

Mean of x_2 => $\mu x_2 = \frac{1+4+2+5+3}{5} = \frac{15}{5} = 3$

Variance $(\sigma^2) = \frac{1}{N} \sum (x_i - \mu)^2$

$x_1 \Rightarrow \sigma^2_{x_1}$

$$\frac{(x_{11} - \mu_{x_1})^2 + (x_{12} - \mu_{x_1})^2 + (x_{13} - \mu_{x_1})^2 + (x_{14} - \mu_{x_1})^2 + (x_{15} - \mu_{x_1})^2}{5}$$

$$\frac{(3-6.6)^2 + (8-6.6)^2 + (5-6.6)^2 + (10-6.6)^2 + (7-6.6)^2}{5}$$

$$\frac{(-3.6)^2 + (1.4)^2 + (-1.6)^2 + (3.4)^2 + (0.4)^2}{5}$$

$$\frac{(12.96) + (1.96) + (2.56) + (11.56) + (0.16)}{5}$$

$\Rightarrow 5.84$

$$\sigma^2_{x_2} = \frac{(x_{21} - \mu_{x_2})^2 + (x_{22} - \mu_{x_2})^2 + (x_{23} - \mu_{x_3})^2 + (x_{24} - \mu_{x_4})^2 + (x_{25} + \mu_{x_5})^2}{5}$$

$$\frac{(1-3)^2+(4-3)^2+(2-3)^2+(5-3)^2+(3-3)^2}{5}$$

$$= (4+1+1+4)/5 = 2$$

$$\sigma^2_{x_1} > \sigma^2_{x_2}$$

$$5.84 > 2$$

Perceptron :

1) Inputs : x_1 & x_2

2) weights associated with : w_1 & w_2

3) Bias $\rightarrow$ Model flexibility (b)

4) Weighted sum

$$z = w_1x_1 + w_2x_2 + b$$

$$z = \sum w_i x_i + b$$

5) Activation function : (Step function)

$$\hat{y} = \begin{cases} 1, & \text{if } z \geq 0 \\ 0, & \text{if } z < 0 \end{cases}$$

6) Updated weight

$$w_i^{New} = w_i^{old} + \eta(y-\hat{y})x_i$$

η = learning rate = 1 (Generally)

Perceptron:

1) Initial

let $w_1 = 0$, $w_2 = 0$, $\eta = 1$

Inputs: $x_1 = 3$, $x_2 = 1$, $b = 1$

$z = 0 \cdot 3 + 0 \cdot 1 + 1 = 1$

$\hat{y} = 1$ but $y = 0$

$\hat{y} \neq y$ prediction fails

$w_1^{New} = 0 + 1(0-1) \cdot 3 = (-3)$

$w_2^{New} = 0 + 1(0-1) \cdot 1 = (-1)$

$w_1 = (-3)$ $w_2 = (-1)$, $x_1 = 8$, $x_2 = 4$, $b = 1$

$z = (-3) \cdot 8 + (-1) \cdot 4 + 1$

$= -24 - 4 + 1 \Rightarrow (-27)$

$\hat{y} = 0 \neq y = 1$

$w_1^{New} = -3 + 1(1-0) \cdot 8 = 5$

$w_2^{New} = -1 + 1(1-0) \cdot 4 = 3$

$x_1 = 5$, $x_2 = 2$, $b = 1$

$z = 25 + 6 + 1 = 32$, $\hat{y} = 1 \neq y = 0$

$w_1^{New} = 5 + 1(1-0) \cdot 5 = 10$

$w_2^{New} = 3 + 1(1-0) \cdot 2 = 5$

$z = (10) \cdot 10 + 5(5) + 1$

$= 100 + 25 + 1$

$= 126$

$\hat{y} = 1$, $y = 1$

$\therefore$ Final weights $= 10$ & 5.